Michigan Clean Water Corps

MiCorps Fact Sheet

Secchi Disk Transparency

What do measurements with a Secchi Disk tell us about a lake?

What is transparency?

Transparency, sometimes referred to as water clarity, is a measure of how far light can travel in water.

What affects transparency?

Transparency is directly affected by the level of suspended particles and dissolved materials in the water.

The main types of suspended particles that affect transparency are algae and sediment. Algae are naturally occurring microscopic plant life found in most water bodies. Algae are a part of a healthy aquatic ecosystem, but their populations can increase dramatically when there are high nutrient inputs into a lake. Too much algae can block sunlight from penetrating deeper into the water, restricting plant growth and altering the balance of the ecosystem.

Suspended sediments can come from many sources. During storm events, high runoff can carry silt and sand from streets, yards, fields and construction sites into aquatic ecosystems. In shallow waters, boats and high winds can stir up bottom sediments. Also in shallow waters, aquatic life, such as carp and crayfish, can stir up the lake sediment. For some lakes, the formation of fine particles of calcium carbonate (marl) in the water is a major factor affecting transparency.

The Secchi disk is named after Pietro Angelo Secchi, who in 1865 used a white disk to take transparency measurements in the Mediterranean Sea (credit: MiCorps).

Dissolved materials in the water can potentially change the water's color (for example, in a "tea" colored lake), which can also affect the water transparency.

Why is transparency important?

Transparency is a basic indicator for the health of an aquatic ecosystem. When transparency is measured consistently week to week and year to year, these measurements are a useful indicator of water quality changes and patterns. When assessed along with other parameters such as chlorophyll *a* and total phosphorus, transparency measurements give us useful insight into the level of biological productivity in a lake, and ultimately its water quality conditions.

How is it measured?

Transparency is most commonly measured using a Secchi disk. This gives a direct measure of how deeply sunlight is penetrating through the water. The Secchi disk is a weighted steel or heavy plastic disk, eight inches in diameter, attached to a line with calibrated measurements.

In the Cooperative Lakes Monitoring Program (CLMP), volunteer monitors measure transparency by lowering the disk down into the water until it is no longer visible, noting the depth on the calibrated line, pulling the disk up until it is visible again, and noting the depth a second time. The Secchi disk transparency is the average of these two depths, rounded to the nearest half foot.

CLMP volunteers measure Secchi disk transparency regularly from mid-May through mid-September. Repeated measurements are necessary throughout the growing season since algal species composition in lakes can change significantly during the spring and summer months, dramatically affect overall water clarity.

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through Michigan Executive Order #2003-15 to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.

Michigan Clean Water Corps

MiCorps Fact Sheet

Chlorophyll *a*

What do chlorophyll measurements tell us about our lakes?

What is chlorophyll *a*?

Chlorophyll is the pigment that allows plants (including algae) to use sunlight to convert simple molecules into organic compounds via the process of photosynthesis. Of the several kinds of chlorophyll, chlorophyll *a* is the predominant type found in green plants and algae. Measuring chlorophyll *a* concentrations in water is a surrogate for actually measuring algae biomass, which is far more expensive and time consuming.

Why is chlorophyll important?

Some amount of algae is naturally present in all healthy lakes. By measuring chlorophyll *a,* we are determining the amount of food available to fuel the lake's food web. Too little chlorophyll *a* indicates that there may not be enough food to support an abundant biological community. On the other hand, too much chlorophyll *a* indicates that nutrient levels in the lake may be artificially high. This is a problem because algae sink to the bottom and decay, a process that depletes deeper water of oxygen. In severe cases, all of the lake's oxygen can be become depleted, resulting in fish kills. In addition, a nutrient-enriched lake with excessive algae is less appealing for recreational activities like boating and swimming.

Bruce Lichliter, CLMP volunteer from Big Glen Lake, prepares to lower the water column composite sampler into the lake (credit: Mike Litch).

What affects chlorophyll?

Algae growth, and thus chlorophyll *a* levels, is directly affected by the amount of nutrients in a lake. Phosphorus is a naturally occurring element in the sediment, water, and biota of a lake. It is the most influential nutrient in most Michigan lakes. Some lakes naturally have more phosphorus than others. However, humans often introduce extra phosphorus into lakes via poorly maintained septic systems and lawn fertilizer, among other sources.

How is it measured?

The amount of chlorophyll in an algal cell varies with algae species as well as with changing light conditions at different depths within the lake. Changing seasons also create different light conditions that, in turn, affect chlorophyll production. To account for some of this variability, Cooperative Lake Monitoring Program (CLMP) volunteers monitor chlorophyll *a* during five sampling events over the summer season (May through September) using a sampling technique that collects water throughout the lake's photic zone. The photic zone of the lake is the upper portion of the water column where sunlight penetrates and supports growth and reproduction of free-floating algae. The CLMP program defines the photic zone as twice the Secchi disk transparency measurement.

Once the water sample is obtained, CLMP volunteers push a known volume of water through a filter disk, which collects the algal cells. The filter disk is frozen and delivered to the Department of Environment Quality laboratory for analysis.

A lake is going to have an amount of variability in the chlorophyll *a* levels over time. Therefore it is very important to monitor it over a long period of time to understand if levels are truly increasing, decreasing, or staying the same. Increasing chlorophyll *a* levels may indicate that management actions are needed to maintain a healthy lake.

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through Michigan Executive Order #2003-15 to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.

Michigan Clean Water Corps

MiCorps Fact Sheet

Total Phosphorus

What do phosphorus levels tell us about a lake?

Why is phosphorus important to lake health?

Phosphorus is an essential nutrient for algae and aquatic plants (which in turn are food for micro-fauna and larger animals). Therefore, phosphorus is an important element of the food chain within a lake. Phosphorus is usually present in very small amounts in a lake and is considered a "limiting factor" for algae and plant growth. That is, even if there is plenty of nitrogen and carbonates, algae and plants growth will be limited by the low amount of phosphorus available.

Lakes with low nutrient levels are said to have a low trophic level–we could say these lakes are "lean" and not very biologically productive; a more technical term would be oligotrophic. Lakes with more, but not excessive, amount of nutrients are called mesotrophic, and lakes over-enriched with nutrients are called eutrophic. Many eutrophic lakes are healthy and have a balanced plant and animals communities. However, highly eutrophic (or "hypereutrophic") lakes can suffer from unsightly algae blooms and are at a greater risk of developing low oxygen conditions because of the decomposition of the excessive plant material.

What is "total" phosphorus?

Phosphorus in water samples can be measured in many different ways. In total phosphorus measurements, chemicals are used to break down all of the solids in the water sample. The final result is a measure of amount of phosphorus in the water and everything contained in the water.

A water sample is properly obtained by lowering the bottle upside down 1-2 feet below the surface, then pointing the bottle up, allowing the bottle to fill (credit: MiCorps staff).

Where does phosphorus come from?

Phosphorus is naturally occurring element in the sediment, water, and biota of a lake. Some lakes naturally have more phosphorus than others. However, humans often introduce extra phosphorus into a lake through poorly maintained septic systems and lawn fertilizer. Other sources in the lake's watershed, such as agriculture, can also contribute. Phosphorus stored in the sediment of a lake can also be released to the water under low oxygen conditions.

How is total phosphorus sampled in the CLMP?

Volunteers in the Cooperative Lakes Monitoring Program (CLMP) simply collect a water sample using a clean water sample bottle provided by the program. The samples are taken from just below the surface of the lake, over the lake's deepest basin. Then, the samples are frozen and delivered to the water quality laboratory of the Michigan Department of Environmental Quality for analysis. Two samples are collected each year: during spring overturn, when the lake is generally well mixed from top to bottom, and during late summer, when the lake is at maximum temperature stratification from the surface to the bottom. Spring overturn is an opportune time of the year to sample just the surface of a lake to obtain a representative sample for estimating the total amount of phosphorus in the lake. A surface sample collected during late summer represents only the upper water layer of the lake, the epilimnion, where most algal productivity occurs. The two taken samples together can help lake managers understand the production dynamics of the lakes and why it may be "behaving" the way it does in terms of algae production.

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through Michigan Executive Order #2003-15 to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.

Michigan Clean Water Corps

MiCorps Fact Sheet

Dissolved Oxygen and Temperature

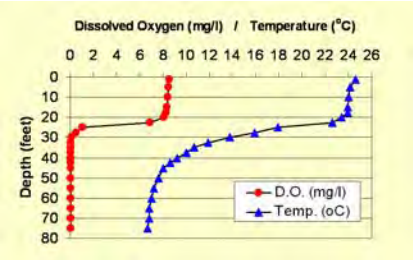
What do DO and water temperature tell us about a lake?

Why are dissolved oxygen and temperature important?

Understanding lake dissolved oxygen (DO) and temperature patterns is important to lake management planning. These patterns reflect and influence lake productivity, and affect lake physical/chemical properties, phosphorus cycling, and fish and aquatic animal populations.

What is DO, and how much does the lake's biology need?

Atmopsheric gases (oxygen, hydrogen, nitrogen, etc) are typically present in water in dissolved form. The amount of DO in water is particularly important, because like humans and land animals, creatures in aquatic environments need oxygen to power the life-sustaining respiration process. Different aquatic creatures require different amounts of DO. For example, trout and stoneflies need DO levels as high as 6-8 mg/L*. Smallmouth bass need DO between 5-6 mg/l. Some animals (like carp) can thrive with very low oxygen (1-2 mg/L, and can even gulp air directly if needed. In any case, when aquatic organisms cannot find water with the proper amount of DO, they will soon die.



This graph of Corey Lake shows the distinct thermal and DO stratification typical for a late summer, good quality, mesotrophic (moderately productive) lake. (credit: MiCorps staff)

What else does DO affect?

DO also affects nutrient cycling in a lake. Low DO in deep water can cause phosphorus to be released from lake sediment. Excessive phosphorus can cause algal blooms. When algae decompose, this causes further DO reductions. This self-fueling pattern of low oxygen, high phosphorus levels, and excessive algae growth is challenging to reverse once it has begun.

How are DO and water temperature connected?

Colder water has the potential to hold more oxygen than warm water. However, colder water doesn't necessarily hold more oxygen in reality. The density of water increases as temperature decreases, so cold water sinks to the bottom of a lake and is separated from the source of oxygen (the atmosphere). Warmer, less-dense water floats at the top of a lake where mixing with the atmosphere keeps the upper waters aerated. The difference in water density results in distinct layers of different temperatures and different DO in a lake. This is known as thermal stratification.

The DO/Temperature meter comes with a very long cable attached to a probe. The probe is lowered from the lake surface to the lake bottom to get DO and temperature profiles (credit: MiCorps staff).

How are DO and temperature measured?

In the Cooperative Lakes Monitoring Program, DO and temperature are measured from the surface to within 3 feet of the bottom, as a profile, in the deepest basin of the lake. Measurements of DO and temperature are made every two to three weeks from mid-May to mid-September. CLMP volunteers use an oxygen/temperature meter connected to a long cable with a sensor probe. The meter displays the DO readout based on the rate of diffusion of molecular oxygen across the probe's membrane. The thermistor component of the instrument provides the temperature readout. Using the DO-temperature data, we can develop dissolved oxygen/ temperature profiles, which show the changes in DO and temperature through depth.

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through Michigan Executive Order #2003-15 to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.

Michigan Clean Water Corps

MiCorps Fact Sheet

Aquatic Plant Identification & Mapping

What plants are in your lake?

Why are aquatic plants important?

Rooted aquatic plants are a natural and essential part of lakes, just as grasses, shrubs and trees are a natural part of the land. Their roots are a fabric for holding sediments in place, reducing erosion and maintaining bottom stability. They provide habitat for fish and invertebrates by providing structure within which to forage, raise young, and hide from predators. Waterfowl, shore birds and aquatic mammals forage on plants, and use them as nesting materials and cover.

Though plants are important to the lake, over-abundant plants can negatively affect fish populations and human recreation. In this situation, it is advantageous to manage the lake and its aquatic plants for the maximum benefit of all users.

To be able to do this effectively, it is necessary to know the plant species present in the lake and their relative abundance and location. A map of a lake showing the plant population locations and densities will greatly aid management projects.

Stacey Daniels samples aquatic plants on Crystal Lake (credit: MiCorps Staff)

What help does the CLMP offer to volunteers interested in aquatic plants?

The Cooperative Lakes Monitoring Program (CLMP) uses qualitative techniques that allow volunteer monitors to generally assess the aquatic plants in their lake. This assessment may be viewed as a "snapshot" of the species of plants in the lake, their general location, and relative abundance. The CLMP assessment provides valuable information about a lake's aquatic plants that is often missing in lake and aquatic plant management programs.

The CLMP provides training and technical assistance to the volunteer monitors enrolled in the aquatic plant survey program. Training in plant identification and mapping is given on an annual basis. Technical assistance in survey design, plant identification, and field technique, including limited on-site consultation, is provided to the volunteer monitors.

What is the procedure for conducting a CLMP aquatic plant survey?

The CLMP uses the procedure written in *A Citizen's Guide for the Identification, Mapping and Management of the Common Rooted Aquatic Plants of Michigan Lakes*. An electronic version of this book can be obtained from www.micorps.net, under *Lake Monitoring* and *CLMP documents*.

In a nutshell, the procedure involves:
1) Using a lake depth map to establish sampling transects.
2) Using a sampling rake to take plant samples at particular depths on each transect.
3) At each sampling point, four samples are taken from the boat at the twelve, three, six, and nine o'clock positions.
4) Identifying the sampled plants and assigning a relative abundance.

Upon completion of the survey, volunteers have a detailed listing of the plant species growing in their lake, including any invasive species that may be problematic. They also will have created a map that shows the overall distribution and density of the lakes' plant population. These products will be valuable in lake management activities and as a reference in the future.

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through Michigan Executive Order #2003-15 to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.

Michigan Clean Water Corps

MiCorps Fact Sheet

Exotic Plant Watch

Are invasive plants threatening your lake?

Aren't plants good for a lake?

Rooted aquatic plants are a natural and essential part of the lake, just as grasses, shrubs and trees are a natural part of the land. However, sometimes a lake is invaded by an aquatic plant species that is not native to Michigan. Some of these exotic plants, like Curly-leaf Pondweed, Eurasian Milfoil, and Hydrilla can be extremely disruptive to the lake's ecosystem and recreational activities.

These exotic plants can "take over" a lake by crowding out beneficial native species. An overabundance of the exotic species can negatively affect fish populations and human recreation.

What steps are needed to prevent exotic plants from taking over my lake?

To avoid having a lake invaded and overrun with exotic plants, the Exotic Plant Watch Program of the Cooperative Lakes Monitoring Program (CLMP) teaches volunteers how to use the five Integrated Pest Management (IPM) strategies:

1. Monitoring

The critical step in stopping the spread of exotic plants is to find them before they spread out across the lake.

Through the Exotic Plant Watch, a plant expert from MiCorps teaches volunteers how to sample their lake for exotic plants and how to identify them. Participants also have the opportunity to mail questionable or unknown plant samples to MiCorps for identification.

In a nutshell, the monitoring procedure involves:

1) Using a lake depth map to establish sampling transects.
2) Using a sampling rake to take plant samples at particular depths on each transect.
3) At each sampling point, four samples are taken from the boat at the twelve, three, six, and nine o'clock positions.
4) Identifying the sampled plants and assigning a relative abundance.

Annual monitoring of the lake for exotic plants can determine if any are present in the lake, and their location and abundance. If monitoring is done frequently and carefully it should be able to detect the first invasion colonies of an exotic plant.

2. Early Detection

Finding the first colonies of an exotic plant invasion will permit a rapid response to control the plant in small areas before it can spread. If the invasion goes undetected in a few years it may cover large areas of the lake.

3. Rapid Response

Responding rapidly to a new invasion of an exotic plant will increase the probability of keeping the plant under control, possibly even eradicating it. Treating small infested areas will be less expensive and more effective than treating large areas after the plant has spread.

4. Maintenance Control

Continually monitoring and treating the exotic plant is time consuming and requires persistence and dedication, but it can keep the plant's population at low, manageable levels. The alternative is to "give up" and let the exotic plant populations expand to cover large areas of the lake, after which control will be very expensive.

5. Preventive Management

Even if the monitoring does not reveal the presence of exotic plants, the lake community should still continue their monitoring efforts. Remember, the key to prevent invasive plant growth is to find them before they take over the lake.

Michigan Clean Water Corps

MiCorps Fact Sheet

Score the Shore: A Shoreline Habitat Assessment

Shoreline habitat is an invaluable component of every lake ecosystem

Why is shoreline habitat important?

Shorelines are the primary habitat area for many of the animals that live on or near a lake. Birds, amphibians, reptiles, insects, and fish all rely on the shallow water and the aquatic and terrestrial vegetation along a lake's edge to find food and a safe place to rear their young. Healthy shorelines also help maintain water quality, limit erosion, and slow and filter rain runoff.

Unfortunately, some Michiganders are loving their lakes to death with excessive lakefront development. The construction of lawns, beaches, and sea walls remove optimal habitat locations along the shoreline. Foot traffic, docks, and the desire for an unobstructed view remove the vegetation that controls erosion.

Understanding where and how the lakeshore habitat is degraded is the first step in alleviating these problems. Lake residents can get this information by going through the Score the Shore process.

How does Score the Shore work?

Volunteers in the Score the Shore assessment use aerial maps and GPS to divide the shoreline of their lake into 1,000 foot sections. Volunteers

Manicured lawns and beaches are often the cause of degraded water quality. (photo credit: MiCorps staff)

then drive a boat around the edge of the lake, filling out an assessment form for each of the 1,000 foot sections. The assessment form contains questions on the number of houses and docks, condition and amount of aquatic and riparian vegetation, and types of erosion control structures.

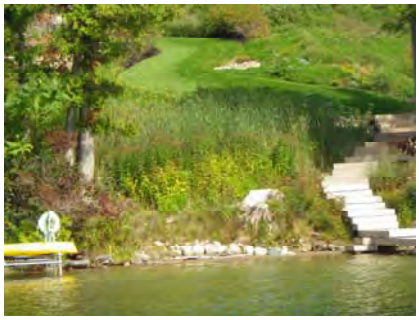
Each of the questions on the form provides a point value, either positive or negative, depending on the condition of the lakeshore. Each 1,000 foot section will have its own score and then an average score is also calculated for the entire lake.

How can the Score the Shore results affect lake management?

With results in hand, lake residents can identify the areas on the lake that are in the poorest condition and why they are in this condition. Residents and lake associations can start both formal educational efforts and informal conversations with neighbors about ways to improve the score. For example, when a shoreline section scores very poorly in woody debris habitat, the lake residents could make efforts to stop pulling out fallen trees or perhaps even place dead trees into the water.

Lake associations, non-profits, and local governments can use their results in writing grant proposals that would provide money for erosion control projects or habitat enhancement.

If you are interested in finding out more information about the Score the Shore program, please visit https://micorps.net/lake-monitoring/.



Lake residents can strike a balance between their needs and the lake's needs by keeping unmowed vegetation between the mowed lawn and the lake. (photo credit: MiCorps staff)

Cooperative Lakes Monitoring Program

The CLMP is the lake monitoring program of the Michigan Clean Water Corps (MiCorps). MiCorps was created through an executive order by Governor Jennifer M. Granholm to assist the Department of Environment, Great Lakes, and Energy in collecting and sharing water quality data for use in water resources management and protection programs. For more information about the MiCorps program, please visit www.micorps.net.